# CHIDUBEM (CHIDU) UMEH

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## SUMMARY

Physicist and quantitative analyst targeting operations research, modeling and simulation, and data science roles in defense and federal settings. Builds statistical models, simulations, and production data pipelines end to end, with a focus on finding where models diverge from observed behavior.

## SKILLS

**Languages:** Python (Pandas, NumPy, Scikit-learn), SQL (PostgreSQL), JavaScript, R, Fortran
**Methods:** Statistical modeling, computational simulation, quantitative risk modeling, data pipeline engineering, data visualization
**Tools:** Tableau, Excel, Git, Jupyter, VASP, LLM assisted development (Claude Code, OpenAI Codex)

## EXPERIENCE

**University of Central Florida, Orlando, FL | Graduate Research and Teaching Assistant** **Jan 2024 to Aug 2026**

- Ran 50+ density functional theory simulations in VASP, modifying Fortran source code, to model defects in hexagonal boron nitride.
- Applied GW and Bethe Salpeter methods to predict optical and electronic properties, then compared predictions against experiment.
- Collected magnetic resonance data across field and frequency sweeps (60 GHz to 1.5 THz, 9 T cryostat) on magnetic crystals.
- Built Python pipelines to validate experimental datasets, flagging where model predictions diverged from measurements.
- Taught and assessed 250+ undergraduate physics students, turning complex concepts into clear, practical explanations.

**Autura | Data Analyst Intern** **Apr 2023 to Aug 2023**

- Joined U.S. Census demographics with internal logistics records in SQL, producing analytics ready datasets for executive reporting.
- Built Tableau dashboards and briefed leadership on findings that informed market expansion decisions.

**Massachusetts Financial Services (MFS), Boston, MA | Information Technology Intern** **Jun 2021 to Aug 2021**

- Delivered a production solution on an enterprise low code platform that extracted data from unstructured PDFs at 90% accuracy.
- Learned the platform quickly and shipped on deadline within a live IT environment at a major asset management firm.

**Bates College, Lewiston, ME | Research Data Analyst** **Mar 2019 to May 2022**

- Analyzed eBOSS spectra and Hubble photometry of starburst galaxies, measuring emission lines to estimate star formation rates.
- Coauthored a peer reviewed publication on galactic outflows and presented the research at the 235th AAS national meeting.

## PROJECTS AND AWARDS

**U.S. Gun Violence Intelligence Tool | Python** **Jan 2026 to Jun 2026**

- Built an end to end pipeline turning 300,000+ CDC WONDER records across 25 years and all 50 states into a deployed tool.
- Delivered race stratified disparity ratios and animated state maps, and extended the dataset into a RAG pipeline demo.

**Erdős Institute Top Project Award | Quantitative Finance Bootcamp** **Nov 2025**

- Built a Black-Scholes-Merton volatility model on the VanEck Africa ETF across 1,000+ trading days.
- Found a 0.27 correlation between implied and realized volatility; selected as a Top Project out of 52 submissions.

**Splittank | Founder and Developer | JavaScript** **Jan 2026 to Present**

- Shipped a full stack trip cost splitting app integrating live EIA and AAA fuel prices, toll calculation, and EV electricity costs.
- Integrated GPS location, FuelEconomy.gov MPG lookups, and payment links across four third party platforms for live users.

## EDUCATION

**University of Central Florida** | M.S. Physics, Concentration in Spintronics **May 2026**

**Bates College** | B.S. Physics, Concentrations in Applying Mathematical Methods and Digital and Computational Studies **May 2022**

## SERVICE

**HeadStart** | Bilingual Literacy Volunteer, weekly sessions; multiple Certificates of Appreciation **Feb 2025 to Jul 2026**